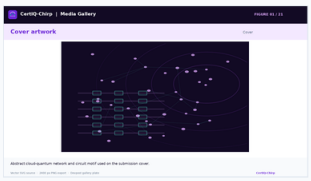
CertiQ-Chirp — Media Gallery

A complete SVG-to-PNG asset pack for the Devpost media gallery. Every diagram originates from a vector SVG, is exported at 2400 px and 1600 px, and is also supplied as a branded 1600×900 gallery plate with a 1200×630 thumbnail.



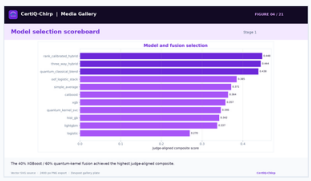
01 Cover artwork



02 End-to-end architecture



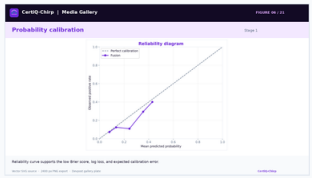
03 Stage 1 leakage controls



04 Model selection scoreboard



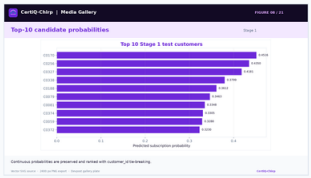
05 Discrimination curves



06 Probability calibration



07 Fusion-weight sensitivity



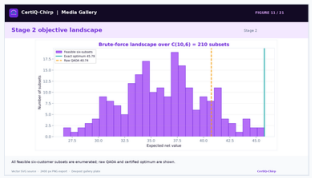
08 Top-10 candidate probabilities



09 Constraint and penalty design



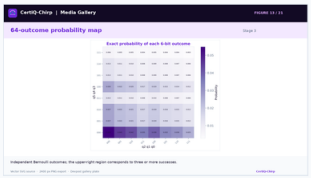
10 QAOA logical circuit



11 Stage 2 objective landscape



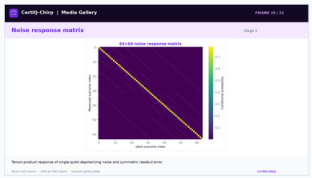
12 Stage 3 product-state circuit



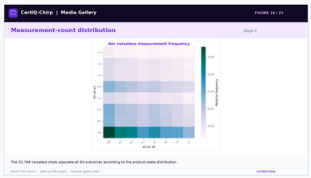
13 64-outcome probability map



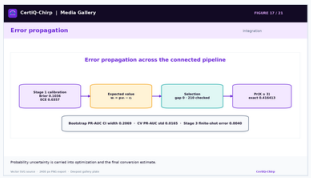
14 Stage 3 estimate comparison



15 Noise response matrix



16 Measurement-count distribution



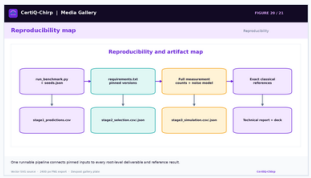
17 Error propagation



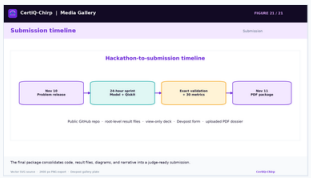
18 Judge rubric readiness



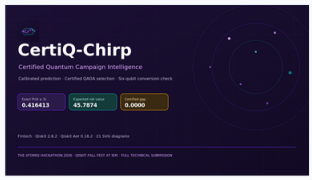
19 Business impact dashboard



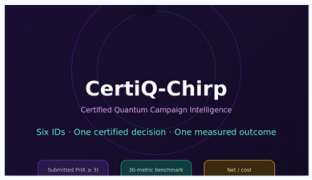
20 Reproducibility map



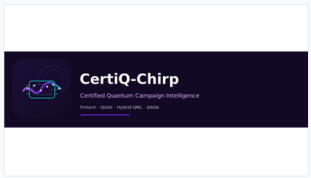
21 Submission timeline



Hero banner



Social card



Wordmark logo

Recommended upload order

1	Hero banner — primary gallery thumbnail
2	Architecture, QAOA circuit, six-qubit circuit, and business dashboard plates
3	Model scoreboard, calibration, Stage 3 estimates, and rubric plates
4	Remaining diagram plates in figure order

Exact values shown across the pack: Pr(K ≥ 3) = 0.416413 exact / 0.420 submitted, expected net value 45.7874, budget 11/25, capacity 16/22, certified Stage 2 gap 0.0000, and 30/30 engineering gates.

Figures 1–6

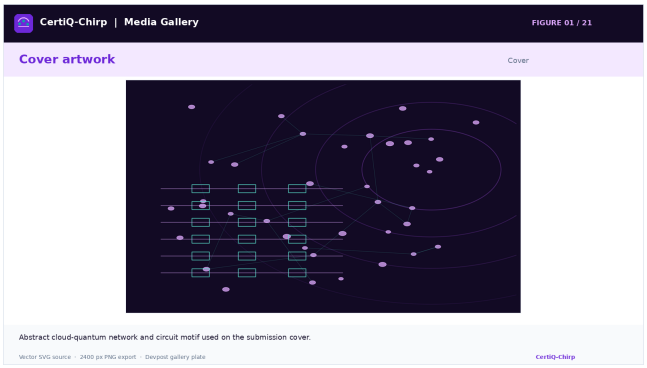


Figure 01 — Cover artwork. Abstract cloud-quantum network and circuit motif used on the submission cover.

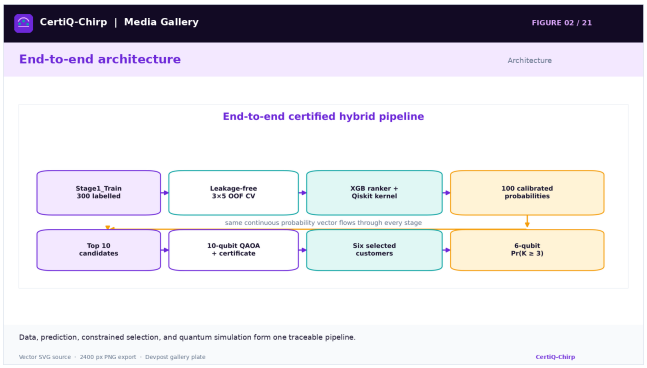


Figure 02 — End-to-end architecture. Data, prediction, constrained selection, and quantum simulation form one traceable pipeline.

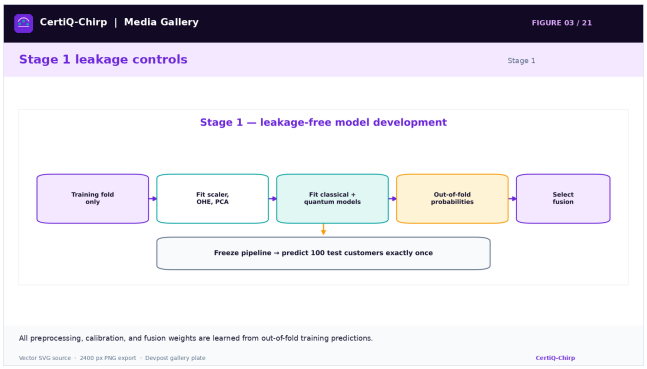


Figure 03 — Stage 1 leakage controls. All preprocessing, calibration, and fusion weights are learned from out-of-fold training predictions.

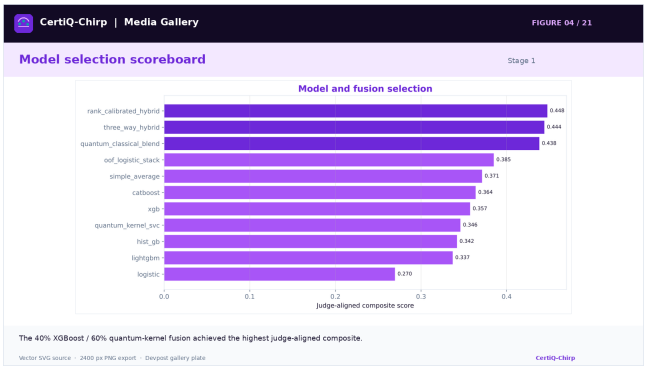


Figure 04 — Model selection scoreboard. The 40% XGBoost / 60% quantum-kernel fusion achieved the highest judge-aligned composite.

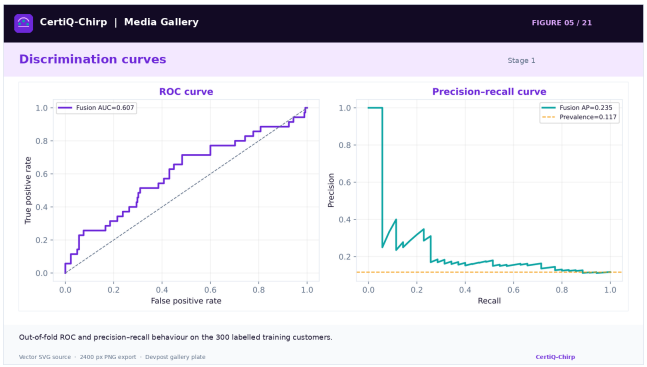


Figure 05 — Discrimination curves. Out-of-fold ROC and precision–recall behaviour on the 300 labelled training customers.

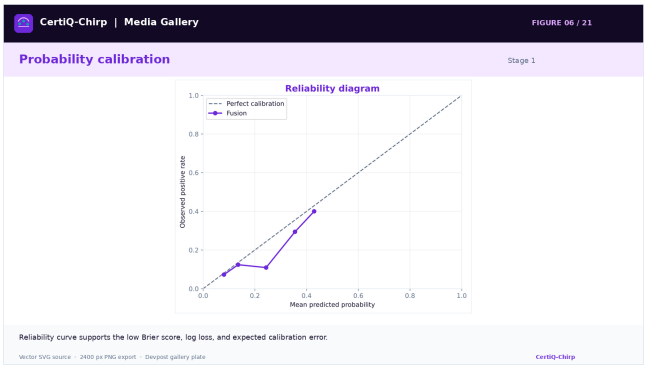


Figure 06 — Probability calibration. Reliability curve supports the low Brier score, log loss, and expected calibration error.

Figures 7–12

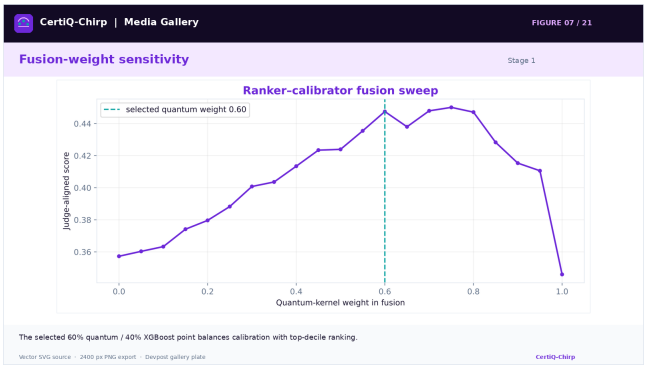


Figure 07 — Fusion-weight sensitivity. The selected 60% quantum / 40% XGBoost point balances calibration with top-decile ranking.

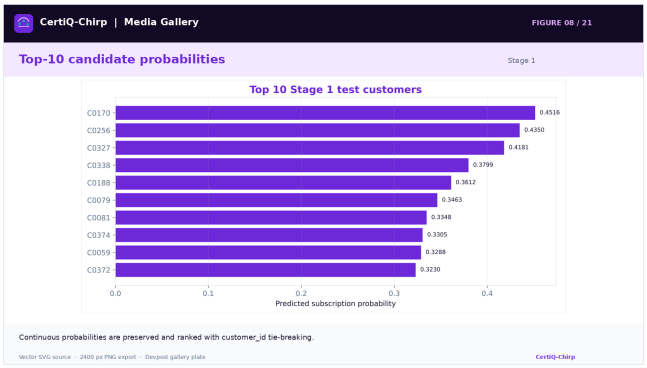


Figure 08 — Top-10 candidate probabilities. Continuous probabilities are preserved and ranked with customer_id tie-breaking.

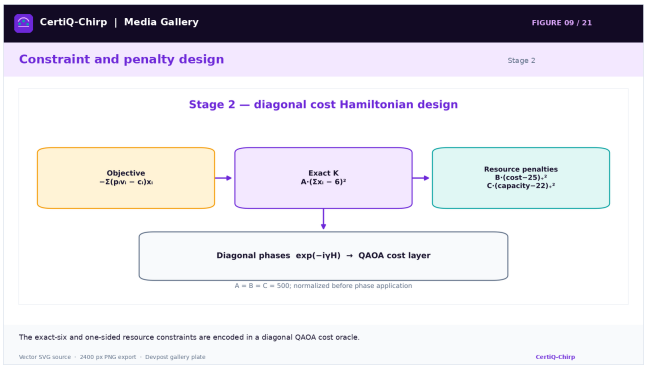


Figure 09 — Constraint and penalty design. The exact-six and one-sided resource constraints are encoded in a diagonal QAOA cost oracle.

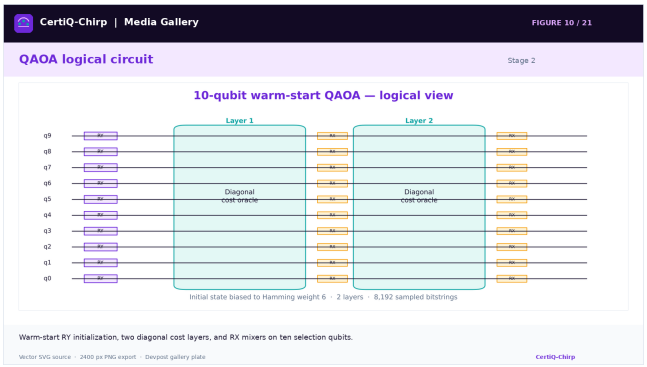


Figure 10 — QAOA logical circuit. Warm-start RY initialization, two diagonal cost layers, and RX mixers on ten selection qubits.

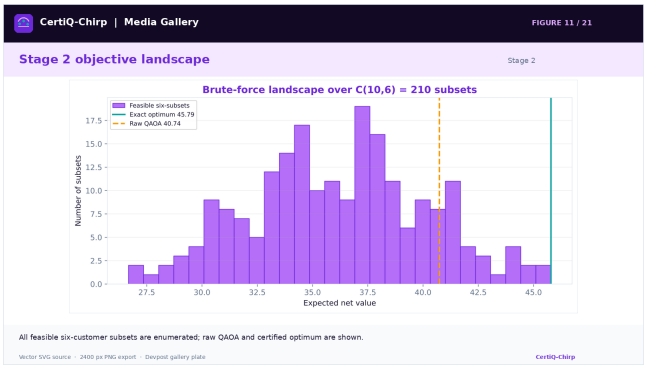


Figure 11 — Stage 2 objective landscape. All feasible six-customer subsets are enumerated; raw QAOA and certified optimum are shown.

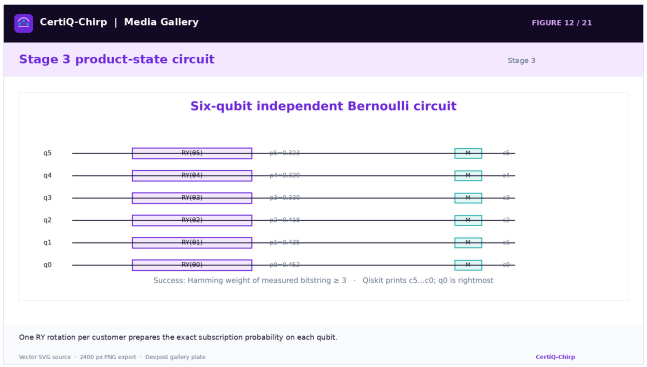


Figure 12 — Stage 3 product-state circuit. One RY rotation per customer prepares the exact subscription probability on each qubit.

Figures 13–18

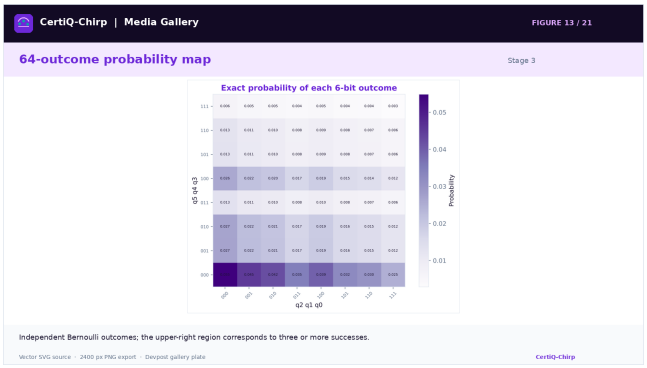


Figure 13 — 64-outcome probability map. Independent Bernoulli outcomes; the upper-right region corresponds to three or more successes.

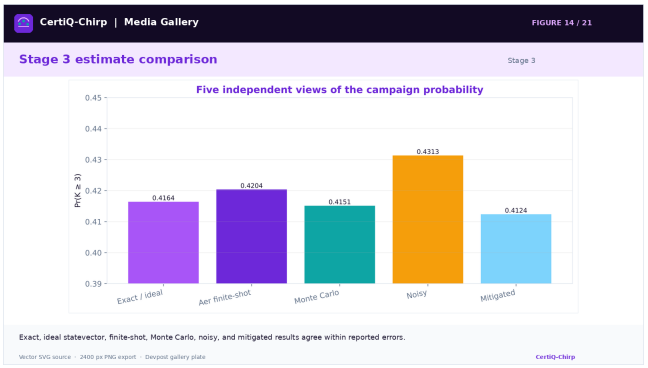


Figure 14 — Stage 3 estimate comparison. Exact, ideal statevector, finite-shot, Monte Carlo, noisy, and mitigated results agree within reported errors.

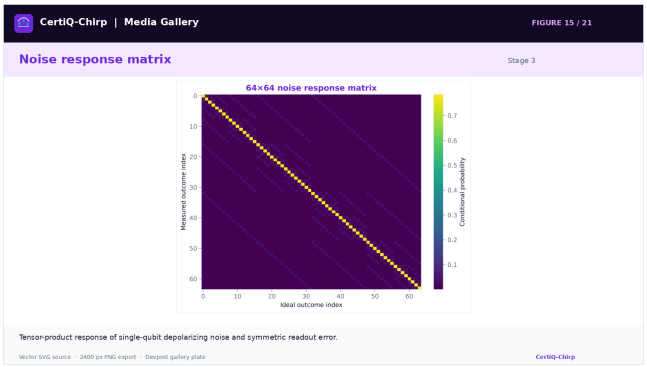


Figure 15 — Noise response matrix. Tensor-product response of single-qubit depolarizing noise and symmetric readout error.



Figure 16 — Measurement-count distribution. The 32,768 noiseless shots populate all 64 outcomes according to the product-state distribution.

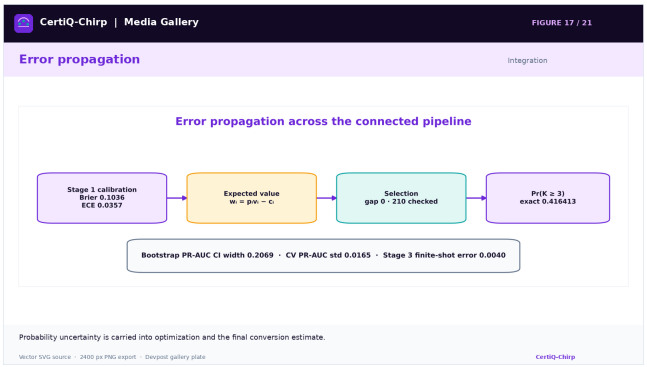


Figure 17 — Error propagation. Probability uncertainty is carried into optimization and the final conversion estimate.

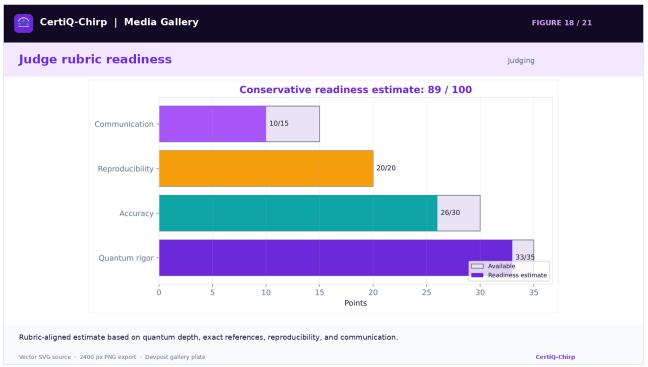


Figure 18 — Judge rubric readiness. Rubric-aligned estimate based on quantum depth, exact references, reproducibility, and communication.

Figures 19–21



Figure 19 — Business impact dashboard. Operational and financial interpretation of the final Stage 2 and Stage 3 results.

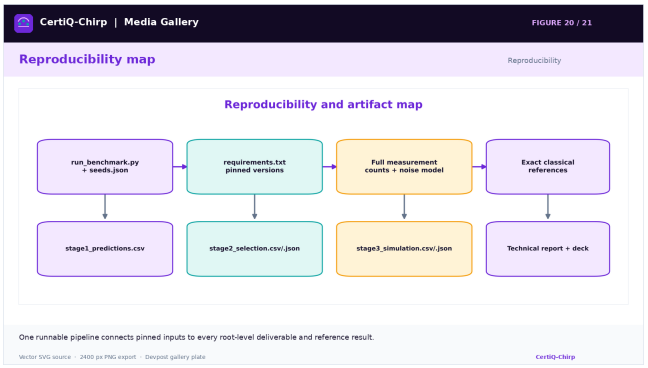


Figure 20 — Reproducibility map. One runnable pipeline connects pinned inputs to every root-level deliverable and reference result.

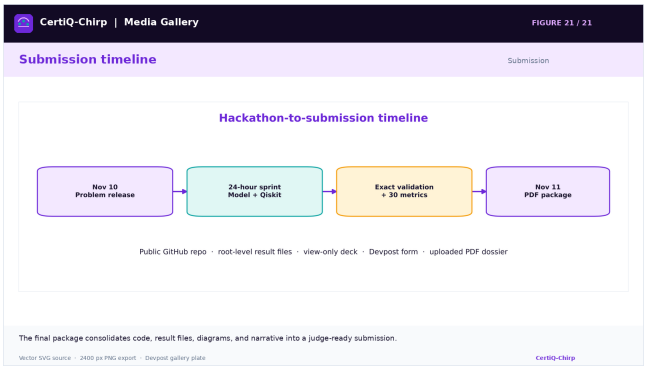


Figure 21 — Submission timeline. The final package consolidates code, result files, diagrams, and narrative into a judge-ready submission.